Higher Math HW2

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1. For each expression, identify every free variable and every bound variable occurrence. Pay attention to variables that appear both free and bound in the same expression.

   Since variable is bound if it lies in the scope of quantifier and is free otherwise:

   a) $\forall x(P(x,y) \Rightarrow Q(x))$;

Occurence of x in $P(x,y)$ is bound. Occurence of x in $Q(x)$ is bound. Occurence of y in $P(x,y)$ is free

b) $(\exists y R(x,y)) \wedge S(y)$;

Occurence of y in $R(x,y)$ is bound. Occurence of x in $R(x,y)$ is free. Occurence of y in $S(y)$ is free

c) $\forall x \exists y(R(x,y) \wedge T(z))$;

Occurence of x and y in $R(x,y)$ is bound. Occurence of z in $T(z)$ is free

d) $(\forall x P(x)) \vee Q(x)$.

Occurence of x in $P(x)$ is bound. Occurence of x in $Q(x)$ is free

2. Let S be a set of biological samples and G a set of genes. Write $D(s,g)$ for the statement "gene g is detected in sample s." Translate each sentence into a formula using quantifiers.

   (a) Every sample has at least one detected gene.
   $\forall s \in S, \exists g \in G | D(s,g)$

   (b) There is a gene detected in every sample.
   $\exists g \in G | \forall s \in S \ D(s,g)$

   (c) Some gene is not detected in any sample.
   $\exists g \in G | \neg(\exists s \in S, D(s,g))$

   (d) There is a sample in which no genes are detected.
   $\exists s \in S | \neg(\exists g \in G, D(s,g))$

3. For the following toy gene-detection table, 1 means "detected" and 0 means "not detected." Let $S = \{s_1, s_2, s_3\}$ and $G = \{g_1, g_2, g_3\}$.

	g_1	g_2	g_3
s_1	1	1	0
s_2	0	1	0
s_3	0	0	1

Decide whether each statement is true or false and briefly justify your answer.

a) $\forall s \in S \exists g \in G : D(s, g)$;
True, as $D(s_1, g_1), D(s_2, g_2), D(s_3, g_3)$

b) $\exists g \in G \forall s \in S : D(s, g)$;
False, as $\forall g \in G, \exists s \in S | \neg D(s, g)$

c) $\forall g \in G \exists s \in S : D(s, g)$;
True, as $D(s_1, g_1), D(s_2, g_2), D(s_3, g_3)$

d) $\exists s \in S \forall g \in G : D(s, g)$.
False, as $\forall s \in S \exists g \in G | \neg D(s, g)$

4. Negate the following statements. Push all negation signs inward, so that they apply only to elementary predicates.

(a) $\forall x \in \mathbb{R} \exists y \in \mathbb{R} : x < y$;
$\exists x \in \mathbb{R} \forall y \in \mathbb{R} : x \geq y$

(b) $\exists x \in \mathbb{R} \forall y \in \mathbb{R} : x \leq y$;
$\forall x \in \mathbb{R} \exists y \in \mathbb{R} : x > y$

(c) $\forall s \in S \exists g \in G : D(s, g)$.
$\exists s \in S \forall g \in G : \neg D(s, g)$

Additionally, determine whether the original statement in (a) is true and whether its negation is true.

Consider $x = n, n \in \mathbb{R}$. Consider $y = n + k, k > 0, k \in \mathbb{R}$. Since $n < n + k$, statement (a) is true. Since statement (a) is true, its negation is false.

5. Consider the set of DNA nucleotides $U = \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\}$ and the subsets

$$X = \{\mathrm{A}, \mathrm{C}\}, \qquad Y = \{\mathrm{A}, \mathrm{C}, \mathrm{G}\}, \qquad Z = \{\mathrm{C}, \mathrm{G}\}.$$

(a) Decide which statements are true: $X \subseteq Y$, $X \subset Y$, $Z \subseteq Y$, $X \subseteq Z$.
$X \subseteq Y$: T;$X \subset Y$: T;$Z \subseteq Y$:T;$X \subseteq Z$:F

(b) List all subsets of Y.
$\{A, C, G\}; \{A, C\}; \{C, G\}; \{A, G\}; \{A\}; \{C\}; \{G\}; \emptyset$

(c) List all strict subsets of Y.
$\{A, C\}; \{C, G\}; \{A, G\}; \{A\}; \{C\}; \{G\}; \emptyset$

(d) Explain why $Y \subseteq Y$ but $Y \not\subset Y$.
$\forall x(x \in Y \iff x \in Y)$, therefore $Y \subseteq Y$
$A \subset B \Rightarrow A \neq B$. Since $Y = Y$, $Y \not\subset Y$

Prove the following statements or provide a counterexample, as requested.

(a) Prove that $A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$.
Since $A \subseteq B$, $\forall x \in A, x \in B$. Since $B \subseteq C, \forall x \in B, x \in C$. Thus, $\forall x \in A, x \in C$. Therefore, $A \subseteq C$

(b) Prove that
$$A \subsetneq B \Leftrightarrow (A \subseteq B) \wedge (\exists x \in B : x \notin A).$$
Consider definition:
$A \subset B \iff A \subseteq B \wedge A \neq B$
$\exists x \in B : x \notin A \Rightarrow A \neq B$. By definition $(A \subseteq B) \wedge (A \neq B) \Rightarrow A \subset B$. Thus RHS implies LHS
$(A \subset B) \iff (\forall x \in A, x \in B) \wedge (A \neq B) \Rightarrow (A \subseteq B) \wedge (\exists x \in B : x \notin A)$. Therefore, LHS implies RHS

(c) Is the following implication always true?
$$A \cup C \subseteq B \cup C \Rightarrow A \subseteq B.$$
If not, construct a counterexample using small finite sets.
Consider $A = \{1, 2\}$
$B = \emptyset$
$C = \{1, 2, 3, 4\}$
$A \cup C = \{1, 2, 3, 4\}$
$B \cup C = \{1, 2, 3, 4\}$
$(A \cup C) \subseteq (B \cup C)$
$A \not\subseteq B$
Thus, the implication is not always true